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Long-term (2000–2023) monitoring and hotspot identification of total suspended solids and turbidity in Lake Tana, Ethiopia, using high temporal resolution MODIS satellite imagery

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Keywords: water quality, Lake Tana, total suspended solids, turbidity

Abstract

Monitoring water quality is challenging in data-scarce regions. This study examines the use of remote sensing to assess key water quality parameters in Lake Tana, Northwestern Ethiopia, to support the sustainable management of Lake water resources. The Moderate Resolution Imaging Spectroradiometer satellite imagery was applied for long-term spatiotemporal assessment of total suspended solids (TSS) and turbidity. The results show that annual mean TSS increased from 350 mg l⁻¹ in 2000–640 mg l⁻¹ in 2020, while turbidity increased from 120 to about 500 NTU before both parameters declined during 2020–2023. Seasonal analysis indicates that the highest TSS and turbidity occurred during the peak rainy month of July. The spatial variation revealed that elevated TSS and turbidity levels were concentrated primarily at the inlets of the major feeder rivers to Lake Tana. This reflects a substantial influx of sediment into the Lake during rainy seasons. During post-rainy months, TSS is redistributed across different sections of the Lake by wind. The findings of this work demonstrate the ability of remote sensing to provide long-term monitoring of water quality parameters, offering valuable insights for the sustainable management of Lake water resources.

1. Introduction

Lakes play a vital role in supporting a wide range of environmental and biological services (Hou *et al* 2017). Globally, they are the sole source of drinking water, irrigation, fisheries, power generation, regional climate regulation, and biodiversity conservation (Woolway *et al* 2020, Makanda *et al* 2022, Mishra 2023). However, the quality of Lake water is progressively deteriorating due to both natural and anthropogenic factors (Shi *et al* 2018, Walton *et al* 2024). Climate change, combined with human activities, is contributing significantly to declines in water quality Solheim *et al* 2010, Li *et al* 2015, Mujere and Moyce 2017). Very rapid population growth, land use/cover changes, agricultural intensification, urbanization, and poor watershed management accelerate the deterioration of Lake water quality, particularly in developing nations (Dodds and Smith 2016). The degradation of quality can be exacerbated by climate variability, including rising temperatures, increased evaporation, and altered precipitation patterns, which affect runoff and sedimentation (Yadava *et al* 2023).

Total suspended solids (TSS) and turbidity are the two primary indicators of water quality. They are essential metrics for tracking water quality and evaluating the extent of pollution (Kusari and Ahmedi 2013, Wu *et al* 2014). These parameters are widely used to track surface water quality, as they respond rapidly to watershed disturbances and can be effectively retrieved from satellite data (Dorji and Fearn 2017, Adjovu *et al* 2023, Friedmann *et al* 2024). Extreme hydroclimate events, such as intense rainfall, can increase soil erosion and sediment transport from upper catchments into the Lake. These events increase TSS and turbidity in the Lake, which affect water quality (Delpa *et al* 2009). Water carries pollutants and suspended sediments/solids, which decrease light penetration through the water column

and affect the aquatic system (Womber *et al* 2021). Elevated TSS and turbidity also increase water treatment costs and pose risks to public health by facilitating the transport of pathogens and contaminants (Zimoch and Paciej 2020).

Lake Tana is the largest Lake in Ethiopia and is used for drinking water supply, irrigation and transportation. The Ethiopian government selected the Lake as a development corridor. However, Lake Tana's water quality is declining due to increased pressure from surrounding catchment land-use changes, agricultural expansion, deforestation, and urban development. Studies have highlighted that Lake Tana's water quality is deteriorating (Goshu and Akoma 2010, Ligdi *et al* 2010, Wondim 2016, Tibebe *et al* 2019, Abegaz *et al* 2023, Engdaw *et al* 2025b). This leads to proliferation of water hyacinth and worse conditions for human and animal health. Many previous water quality studies in Lake Tana (e.g. Goshu *et al* 2017, Tibebe *et al* 2019, Womber *et al* 2021) primarily relied on grab sampling, focusing on localized areas around towns and at the confluence of a single river. Their findings indicated increased pollutant accumulation in the Lake, attributed to elevated nutrient inflows, which have negatively impacted water quality. However, the limited geographic scope of these studies did not provide a comprehensive assessment of Lake Tana as a whole. Moreover, the field sampling analysis is limited to temporal variation and does not account for the long-term dynamics of TSS and turbidity. Those gaps highlight the need for more extensive spatiotemporal assessment (Zhao *et al* 2020, Yunus *et al* 2021). Thus, an effective alternative is to use satellite datasets, which can reduce sample size and effectively address spatial and temporal variations (Glasgow *et al* 2004). Moderate Resolution Imaging Spectroradiometer (MODIS) imagery has been widely used for long-term monitoring of inland water quality due to its high temporal resolution and consistent data coverage within 8 d (Tarrant *et al* 2010, Shi *et al* 2018, Cao *et al* 2023). It has demonstrated the capability to estimate TSS and turbidity at regional and global scales for open water bodies, particularly for Lakes (Chen *et al* 2015, Rocha *et al* 2024).

Therefore, this research aimed to (i) examine the long-term spatio-temporal trends of TSS and turbidity (2000–2023) and (ii) delineate the hotspot sites highly sensitive to TSS and turbidity. The study findings will serve as guidance for decision-making and implementation of sustainable management of Lake Tana and similar Lake water resources globally. The results are expected to support extension workers, water resource managers, and environmental planners involved in Lake water resource planning and management activities.

2. Materials and methods

2.1. Study site

Lake Tana is the largest Lake in Ethiopia, located in the northwestern highlands, and serves as the primary source for the Blue Nile River. The Lake covers an area of approximately 3000 km². It is geographically situated in the 37 UTM zone between 12°0' latitude and 37°15' longitude (figure 1). The Lake drains a wide catchment area (Tana Sub-Basin) and has an average depth of 9 m. Many permanent rivers, including Gilgel Abay, Gumara, Ribb, Megech, and small tributaries, feed the Lake. The Tana Lake water is utilized for drinking, hydropower, agriculture, fishing, tourism, and transportation. The government established a growth corridor with many irrigation schemes around Lake Tana and its surrounding watersheds. The catchments of the Lake Basin experience a unimodal rainfall regime with peak precipitation during Kiremt (Ethiopian rainy season) from June to August.

2.2. Data sources and acquisition for monitoring TSS and turbidity

MODIS provides long-term, continuous Earth observation data that are particularly useful for large inland water bodies due to its high temporal resolution and radiometric stability (Zhou *et al* 2018, Xiong *et al* 2020). The current study utilized the multitemporal MODIS Terra Surface Reflectance product (MOD09A1) to estimate the TSS and turbidity levels over Lake Tana. It was applied to analyze the reflectance values in specific spectral bands using the Google Earth Engine platform (GEE). The GEE cloud-computing environment enables efficient handling of large time-series datasets and supports reproducible analysis of long-term spatiotemporal dynamics for environmental assessments (Zhao *et al* 2021, Ghosh *et al* 2022, Chen *et al* 2024). MODIS-Terra images have been obtained at a spatial resolution of 250 m and a temporal resolution of 8 d. The images were preprocessed, and near-infrared (NIR) reflectance values were extracted from single bands. Quality assurance layers were applied to mask clouds and poor-quality pixels to reduce noise and uncertainty in the reflectance data.

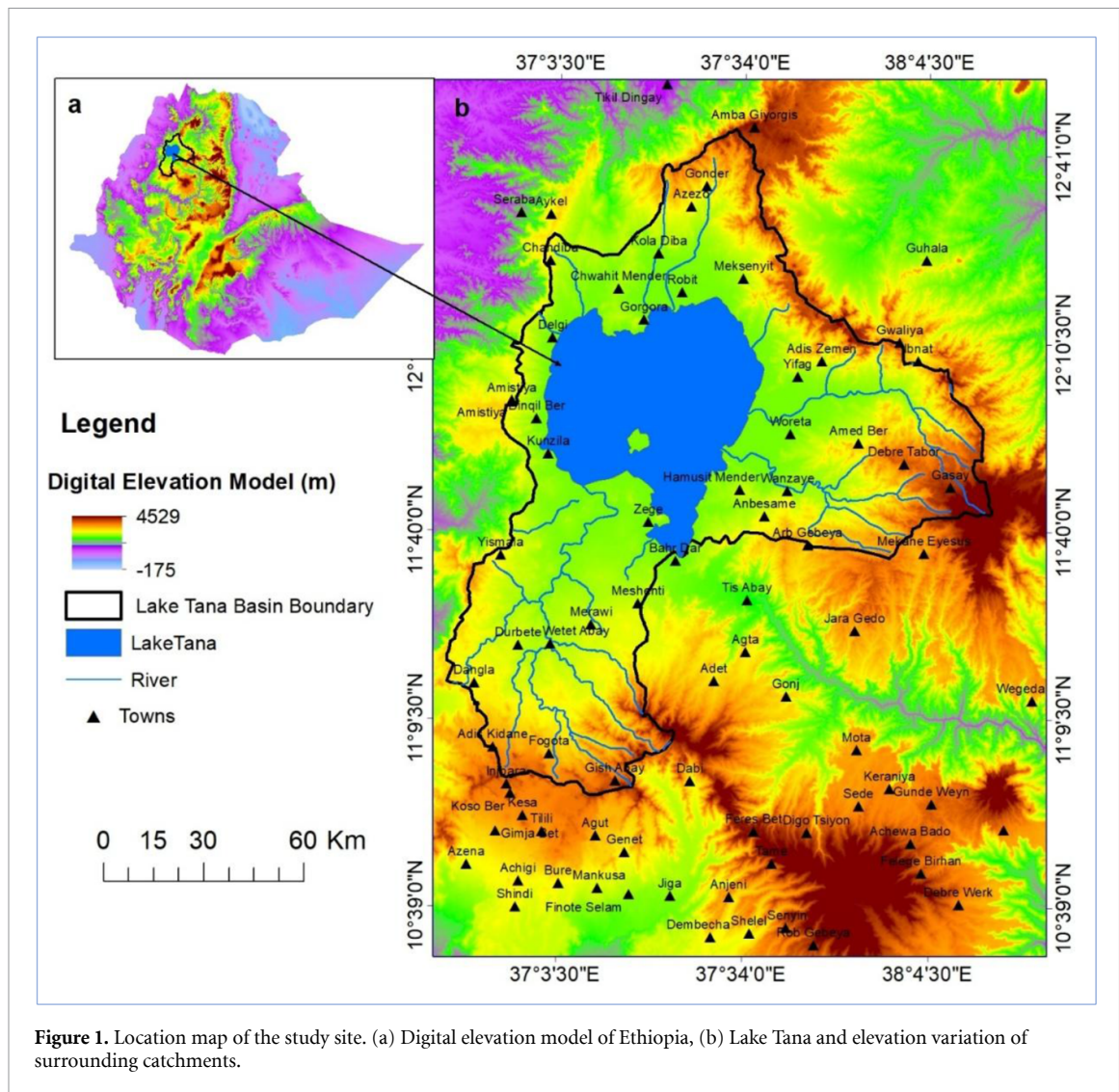


Figure 1. Location map of the study site. (a) Digital elevation model of Ethiopia, (b) Lake Tana and elevation variation of surrounding catchments.

Atmospheric correction was applied to the MODIS images to remove atmospheric interference and obtain accurate surface reflectance values. This step is crucial for reducing the effects of aerosol scattering and gaseous absorption, thereby improving the reliability of satellite-derived water quality parameters (Vermote and Kotchenova 2010). Only high-quality water pixels were retained for analysis to minimize uncertainties associated with atmospheric contamination and mixed pixels. These preprocessing steps substantially reduced noise and improved the reliability of the retrieved water quality parameters.

2.3. Retrieval of TSS and turbidity

Suspended sediments strongly influence the spectral reflectance of inland waters, particularly in the NIR region. In the NIR, water absorption is dominant, whereas scattering is primarily controlled by suspended particulate matter. Consequently, NIR surface reflectance is expected to exhibit an approximately linear relationship with suspended sediment concentration. This relationship has been widely validated in previous studies and forms the theoretical basis for satellite-based retrieval of TSS and turbidity in optically active inland waters (Sutherland *et al* 2000, Doxaran *et al* 2002).

In the current study, GEE scripts were developed to extract surface reflectance values from the MODIS NIR band for each image acquired during the study period (2000–2023). The extracted NIR surface reflectance values were subsequently converted to TSS and turbidity using previously developed and validated empirical regression models for Lake Tana. These models were established by relating *in situ* measurements of TSS and turbidity reported in earlier studies (Goshu *et al* 2017, Tibebe *et al* 2019, Womber *et al* 2021, Leggesse *et al* 2024, Nkwasa *et al* 2024) to the corresponding MODIS surface reflectance values. The resulting long-term time series enabled the assessment of monthly and annual

variations in TSS and turbidity, facilitating the identification of seasonal dynamics, interannual trends, and persistent suspended sediment hotspots across Lake Tana and its major river inflows.

2.4. Temporal trend, seasonal analysis and hotspot identification

Monthly and annual composites were generated from all valid MODIS observations between 2000 and 2023. Monthly mean TSS and turbidity maps were produced by averaging all cloud-free observations within each calendar month. Annual mean values were then calculated from monthly composites to investigate long-term temporal variability. The long-term temporal trends of TSS and turbidity were evaluated using annual averages. Seasonal variability was examined using mean monthly composites. Differences between wet and dry seasons were also investigated to understand the influence of seasonal rainfall on sediment transport dynamics. Spatial distribution maps of TSS and turbidity were produced using long-term monthly averages derived from MODIS. These maps enabled identification of spatial variations across Lake Tana throughout the study period. Hotspot areas were identified that consistently exhibited elevated TSS and turbidity concentrations. Persistent hotspot regions were delineated by comparing long-term mean monthly spatial maps and identifying areas with recurrent high sediment concentrations.

2.5. Validation of the retrieved TSS and turbidity

Field measurements of TSS and turbidity were not conducted in the current study. Consequently, indirect validation was performed using three complementary approaches. First, the spatial and seasonal patterns of MODIS-derived TSS and turbidity were compared with previously published field investigations (Goshu *et al* 2017, Tibebe *et al* 2019, Womber *et al* 2021, Leggesse *et al* 2024, Nkwasa *et al* 2024) conducted in Lake Tana. Second, the empirical relationships used to estimate TSS and turbidity have been extensively validated in previous studies of inland lakes, where coefficients of determination (R^2) typically range from 0.65 to 0.85 (Doxaran *et al* 2002, Shi *et al* 2018). Finally, the physical consistency of the retrieved TSS and turbidity values was assessed by comparing them with long-term monthly and annual rainfall records. Increased rainfall is expected to enhance soil erosion and sediment transport into the lake, resulting in elevated suspended sediment concentrations. Agreement between rainfall variability and the derived TSS and turbidity, therefore, provides additional confidence in the retrieved products.

3. Results and discussion

3.1. Mean monthly and annual TSS and turbidity variations (2000–2023)

MODIS satellite images processed with the GEE platform were used to assess surface reflectance over Lake Tana. Reflectance values were used to calculate TSS and turbidity for more than 2 decades, from 2000 to 2023, due to the scarcity of field-measurement data. The use of these long-term satellite observations enabled consistent monitoring of sediment dynamics and water turbidity across the entire surface of Lake Tana. This overcomes the temporal and spatial limitations of conventional field-based sampling, and no sampling was conducted in the current study. The results show an increasing annual trend in TSS from 2000 to 2020 (figure 2(b)) and a somewhat declining trend over the three years (2021, 2022, and 2023). This long-term increasing trend suggests a gradual intensification of sediment inputs to the Lake. The sediment entering the Lake is due to land/use changes, agricultural expansion, and soil degradation across the Lake Tana Sub-Basin over time. Turbidity has the same trend as TSS, increasing from 2000 to 2020. The TSS concentration could increase turbidity levels. At shallow depths, sediment resuspension further elevated turbidity, particularly during post-rainy seasons (Kebedew *et al* 2020). Wind and wave action in the shallow part of the Lake reintroduce bottom sediment into the Lake's water column. This further increases the lake's turbidity. A general upward trend shifted in 2021, with a concerning decline in TSS values that persisted through 2023, as illustrated in figure 2(b). The recent annual decline in TSS and turbidity levels may be due to decreased rainfall intensity and effective watershed management practices that reduce sediment erosion entering the Lake. The recent Ethiopian green legacy (since 2019) may improve watershed management and reduce soil erosion. Ethiopia's massive reforestation campaigns aim to reduce soil erosion in the Upper Blue Nile Basin (Beyene and Shumetie 2023).

Seasonal analysis reveals that TSS concentrations and turbidity levels are significantly elevated during the rainy seasons, particularly in June, July, and August (figure 3(a)). This is related to higher sediment loading from nonpoint sources in intensively agricultural watersheds. Rainy seasons transport soil nutrients into the Lake from surrounding catchment areas (Setegn *et al* 2008, Taffese *et al* 2014). This seasonal variation highlights the critical impacts of precipitation on sediment transport mechanisms and

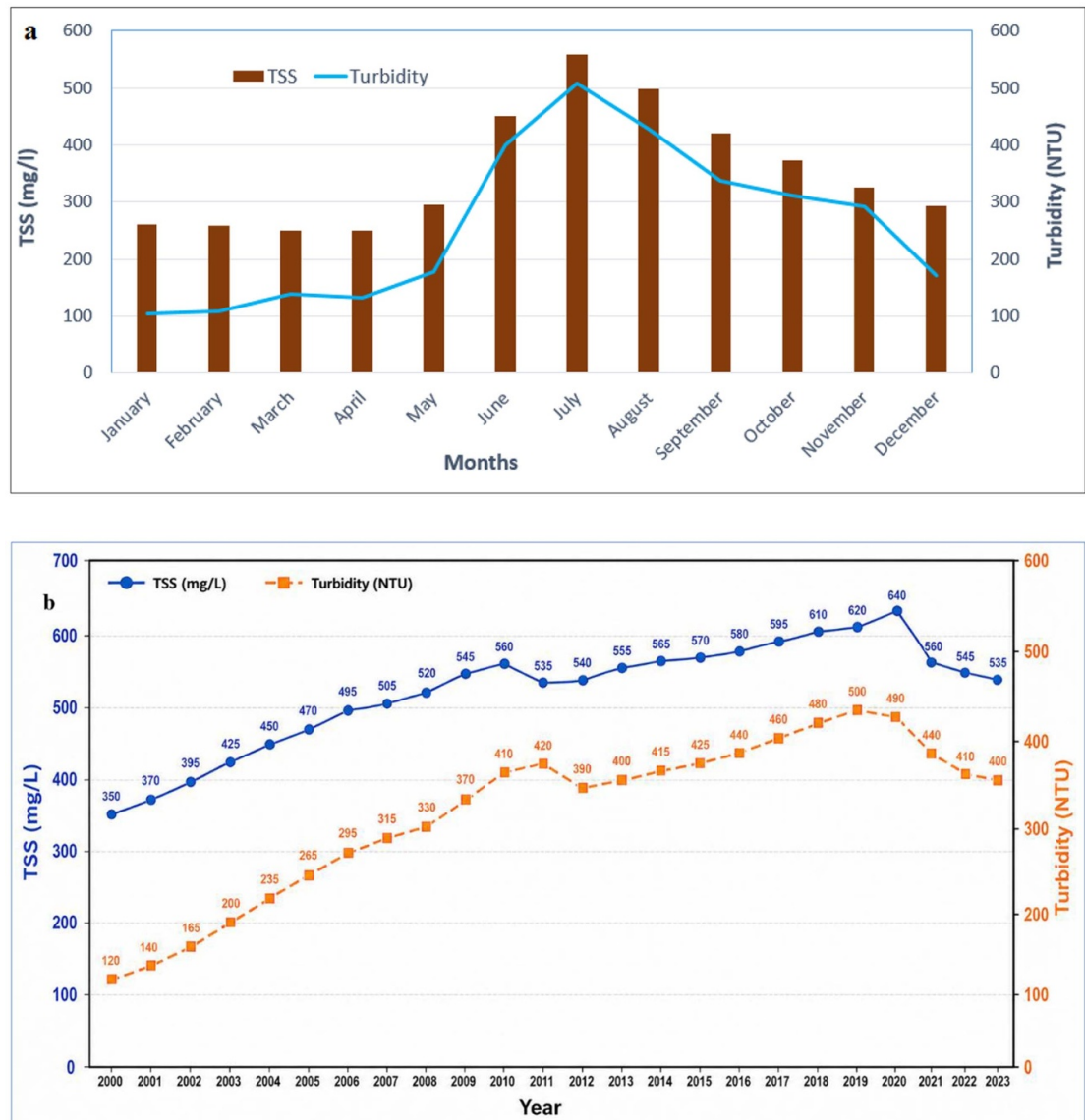


Figure 2. (a) Mean monthly historical TSS and turbidity, (b) annual TSS and turbidity derived from MODIS image surface reflectance.

water quality deterioration within the Lake. Previous studies showed that peaks in TSS and turbidity correspond to intense rainfall and elevated TSS levels observed during rainy seasons (e.g. Setegn *et al* 2008, Taffese *et al* 2014, Lemma *et al* 2018). This is because surface runoff enhances sediment transport to the Lake. Recent studies (e.g. Womber *et al* 2021, Leggesse *et al* 2024, Engdaw *et al* 2025b) also confirm that the highest turbidity levels coincide with the rainy seasons. This is because of the high sediment input to the Lake from the major feeder rivers, Gilgel Aaby, Gumara, Ribb, and Megech. Those studies can validate the current satellite-derived estimates from MODIS images. During the post-rainy months from September to November, TSS and turbidity remain elevated, and scholars (e.g. Kebedew *et al* 2020) suggest this is due to sediment resuspension driven by wind speeds. The minimum TSS concentration and maximum water clarity (i.e. minimum turbidity) were observed during the dry seasons from January to April. However, the agricultural recession along the shores of Lake Tana causes sediment to enter the lake, elevating TSS and turbidity during these months. The expansion of water hyacinth traps fine sediments and can lower TSS concentrations in parts of the Lake where it has expanded (Kebedew *et al* 2023).

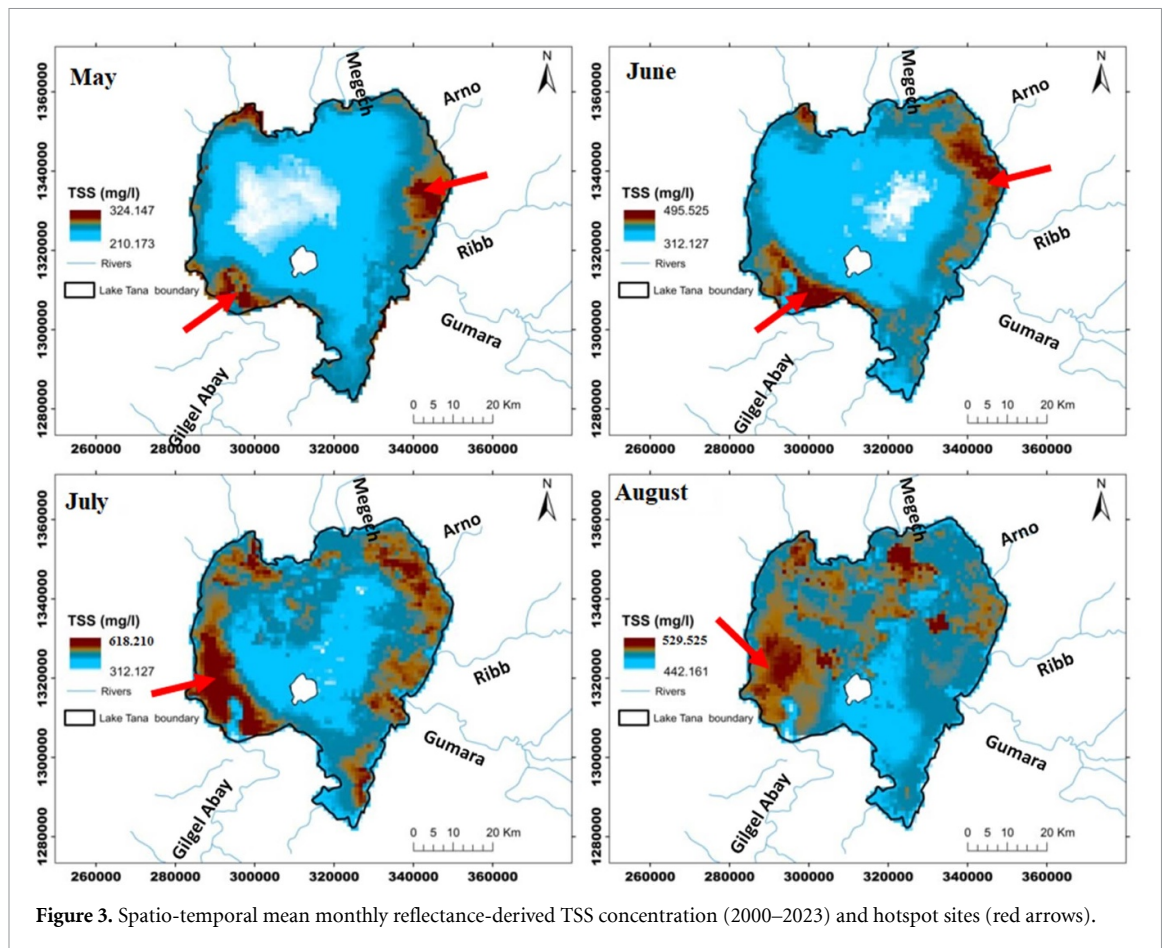


Figure 3. Spatio-temporal mean monthly reflectance-derived TSS concentration (2000–2023) and hotspot sites (red arrows).

3.2. Spatial variation of TSS and turbidity and hotspot sites

The monthly mean spatial MODIS-derived TSS concentration is shown in figure 3. The spatial patterns are mainly influenced by the Lake's feeder rivers. The section of the Lake surrounding the inlets of Gilgel Abay, Gumara, Ribb, Megech, Arno, and other small tributaries produced anomalies characterized by elevated TSS levels (figure 3). The spatial distribution maps for June, July, and August show elevated TSS hotspots at the river inlets of major feeder rivers to the Lake. This indicates strong hydrological connectivity between the catchment and the Lake. The result aligns with studies by Lemma *et al* (2018) and Zimale *et al* (2018), indicating that sediments are transported to the Lake and significantly increase TSS concentrations at the inlets during rainy seasons. The Gilgel Abay River is the major sediment-inflow path to the Lake and is the main hotspot region (figure 3). According to Kebedew *et al* (2021), 60% of the inflow comes from this river. Hence, it carries a large amount of sediment to the Lake and has the dominant role in sediment delivery during high-flow conditions. The high concentration of TSS around the Gilgel Abay River may be due to the steep-slope geomorphology, which promotes rapid runoff into the Lake. Additionally, the limited floodplain development at the Gilgel Abay outlet reduces sediment deposition before joining the Lake. Elevated, steep-slope regions are prone to soil erosion, contribute significantly to sediment loads, and increase TSS in aquatic environments (Lewin *et al* 2017, Wohl 2020). The sediments cannot be deposited on land near the Gilgel Abay River inlet. This results in direct sediment entering the Lake water column. Although it is the largest supplier, its sediment typically spends less time near the inlet and migrates more quickly toward its outlet in the Blue Nile River (Abay River) at the southern Gulf. Womber *et al* (2021) describes a highly mobile river that generates a sustained plume of high turbidity in the southern Gulf. Similarly, in the current study, the southern part of the Lake at the outlet of the Abay River showed elevated TSS and turbidity, indicating sediment transport to the lower catchments and to lower Nile River countries.

The eastern and northern borders of Lake Tana, particularly the Dembia floodplain (influenced by the Megech River) and the Fogera floodplain in the east (primarily impacted by the Gumara and Rib

rivers), experience significant flooding during the rainy seasons. As a result of these seasonal floods, a substantial amount of sediment and suspended materials is deposited onto the floodplain lands and the Delta development, rather than being carried into the Lake. This process results in lower sediment input to the Lake from the Gumara and Ribb rivers than from the Gilgel Aaby river. The development of Deltas increased over time with the entry of the Gumara (Abate *et al* 2015). These sediment dynamics explain the lower TSS estimate in the Gumara and Ribb River inlets compared to the Gilgel Abay River inlet. During peak flow, the floodplains absorb excess water and facilitate sediment deposition before the water reaches the Lake. Thereby reducing the overall TSS levels in the Lake water column in these regions. Kebedew *et al* (2023) applied hydrodynamic modeling to demonstrate that the Lake current generally transports sediment and nutrients from the eastern river inlets (Rib and Gumara) towards the northeast. In the central part of the Lake, the TSS concentration is minimal across all months. This is due to a lack of sediment sources and being far from the feeder rivers. The sediments carried by feeder rivers settle before reaching those parts of the Lake.

Wind plays a role in sediment distribution and spatial dynamics in Lakes (Hofmann *et al* 2011, Nutz *et al* 2015). Wind-generated waves can resuspend settled sediments in the lake's water column (Larsen and MacDonald 1993, Hofmann *et al* 2011). This refers to wind-induced sediment resuspension, especially in shallow Lakes where bottom sediments are easily reworked by wave action. In Lake Tana, winds affect sediment distribution, particularly pronounced during the windy months (post-rainy months) of September, October, November, and December, when strong winds are more frequent. During these post-rainy periods, characterized by reduced water depth and exposed shallow zones, the susceptibility of bottom sediments to resuspension increases. The sediment particles are lifted and carried in the direction of the wind (Engida 2010, Kebedew *et al* 2020). This enhances lateral sediment transport across the Lake surface. The sediments are typically redistributed toward the northeast and east (figure 5, yellow arrows) and are consistent with the work of Kebedew *et al* (2020). This pattern clearly shows wind direction over the Lake during windy dry months and post-rainy seasons. As a consequence, sediments tend to accumulate more significantly in the north and eastern regions of the Lake (figure 5, yellow arrows).

Most of the turbidity is caused by high TSS entering the lake. In Lake Tana, turbidity is caused by both natural and anthropogenic factors (Goshu *et al* 2017, Tibebe *et al* 2019). The high level of TSS is mainly associated with sediment inflow to the Lake. TSS levels determine water clarity. Runoff carries weathered rock debris and soil erosion from agricultural areas into the Lake. This increases the level of turbidity. Turbidity varies across time and space because it depends on factors such as precipitation amounts, hydrological characteristics, landscapes, and seasons (Goshu *et al* 2017, Tibebe *et al* 2019). Steep-slope and mountainous areas are more prone to erosion during rainfall, leading to higher turbidity in those sections of the Lake than in flatter regions. The western parts of the Lake, where steep catchment topography generates high sediment transport to the Lake, are directly affected. The area of Lake Tana near the Gilgel Abay River inlet consistently exhibits higher turbidity (figure 5), as evidenced by elevated TSS (figure 4). This is a direct consequence of the river's sediment load, particularly during rainy seasons when runoff increases intensely. As shown in figure 5, high turbidity values have been observed from June to August. This is because those months are rainy and load the lake with high sediment. Spatially, the major river inlets to the Lake are characterized by high turbidity. In particular, in the western part of the Gilgel Abay River inlet, highly elevated turbidity is observed, similar to TSS levels. The eastern parts show a lower level than the western part because, in the eastern part, sediments are deposited on gentle slopes (e.g. the Fogera floodplain), and the level of sediment joining the Lake is lower than in the southwestern Gulf.

3.3. Validation of the MODIS-derived TSS and turbidity values

Long-term *in situ* measurements of TSS and turbidity are scarce over Lake Tana. Direct point-by-point validation of the current MODIS-derived values was constrained. Therefore, the current study values of TSS and turbidity are validated by previous field measurement reports (previous studies), the coefficient of determination values (R^2) between NIR reflectance-derived and field-based TSS and turbidity measurements, and the correlation of monthly and annual rainfall trends with current MODIS-derived TSS and turbidity levels. First, the current TSS and turbidity values are validated against previous studies using available measurements. The seasonal and spatial patterns were compared with those reported in previous studies. As a result, the current MODIS-derived values are highly consistent with field-based analyses reported by Goshu *et al* (2017), Tibebe *et al* (2019) and Womber *et al* (2021) (table 1). In these

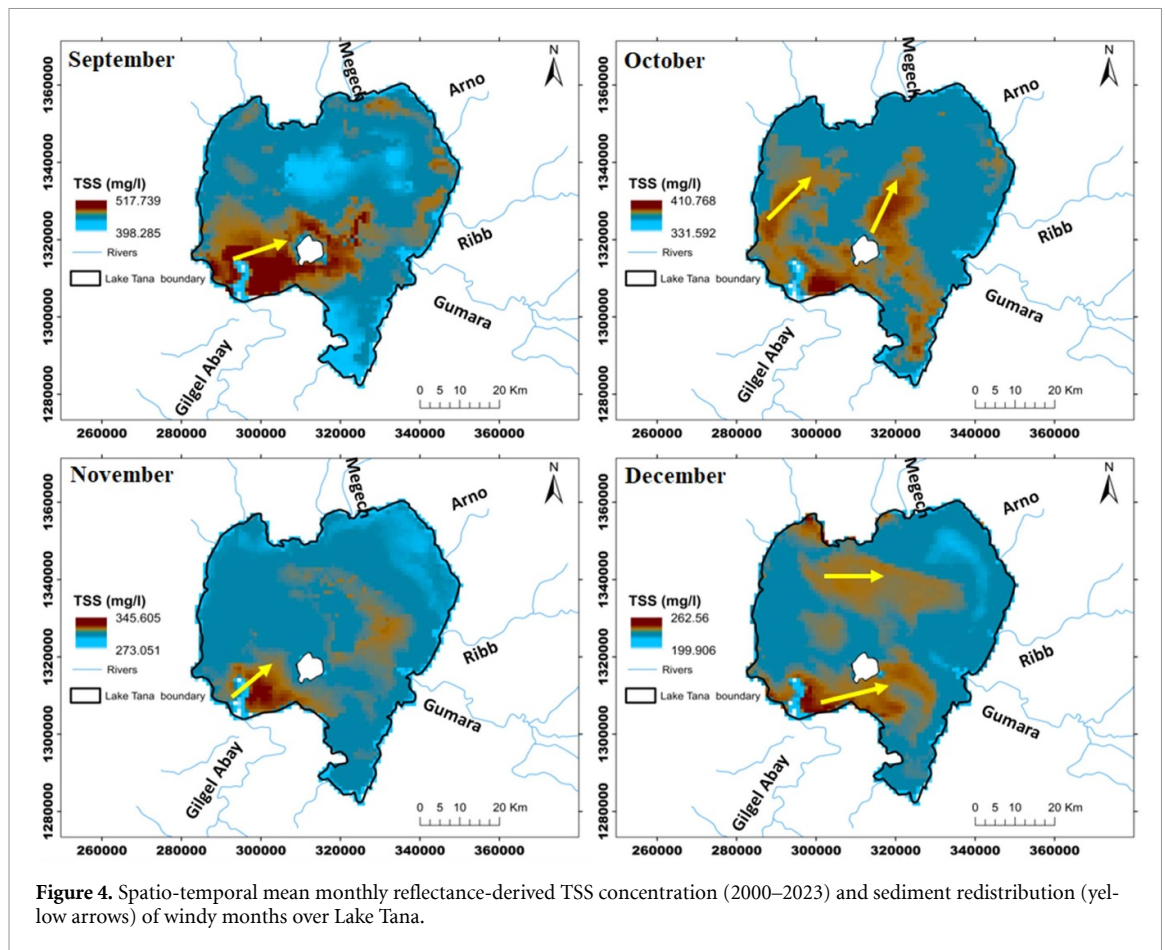


Figure 4. Spatio-temporal mean monthly reflectance-derived TSS concentration (2000–2023) and sediment redistribution (yellow arrows) of windy months over Lake Tana.

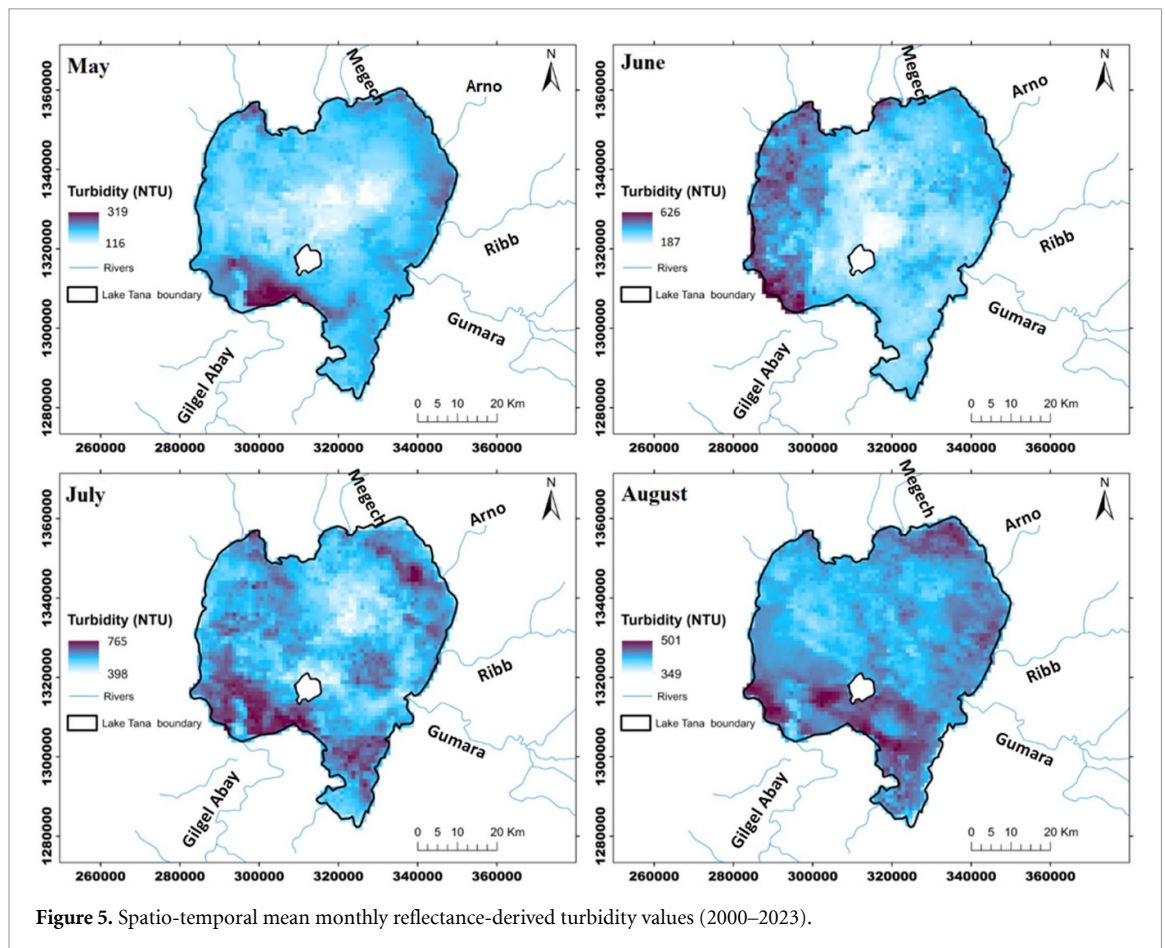
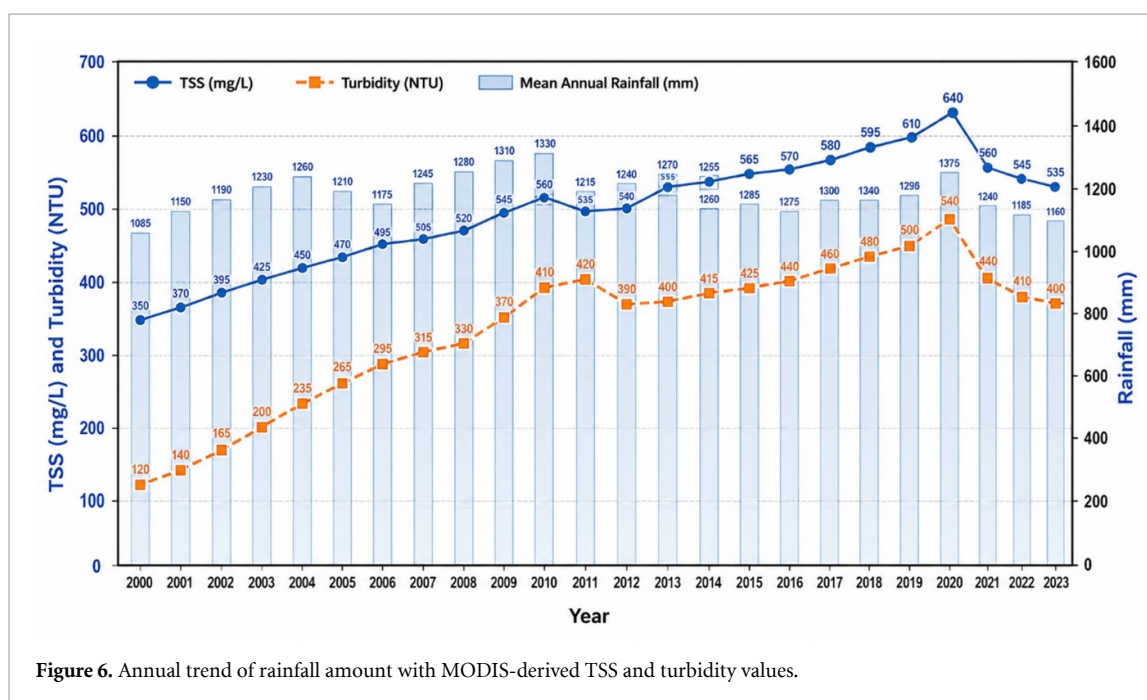


Figure 5. Spatio-temporal mean monthly reflectance-derived turbidity values (2000–2023).

Table 1. Validation of the current MODIS-derived TSS and turbidity levels with previous field measurement reports and studies.

Previous study	Method	Location	Season	Field observation (previous studies)	MODIS-derived levels (current study)
Tibebe <i>et al</i> (2019)	Grab sampling	Near Bahir Dar Town	Wet	High turbidity	High TSS and turbidity
Womber <i>et al</i> (2021)	RS and field sampling	Gilgel Aaby River inlet	Wet	Peak TSS	Peak TSS and turbidity
Goshu <i>et al</i> (2017)	Field measurement	Lake-wide	Dry	Low TSS level	Low TSS and turbidity
Leggesse <i>et al</i> (2024)	Machin learning and RS (Sentinel)	Southern gulf	Wet/Post-wet	Elevated turbidity plumes	Elevated turbidity
Nkwasa <i>et al</i> (2024)	SWAT+ and Remote Sensing	Ribb and Gumara inlets	Wet	High sediment loading	High TSS hotspots

**Figure 6.** Annual trend of rainfall amount with MODIS-derived TSS and turbidity values.

studies, elevated concentrations of suspended sediment and turbidity were observed during the wet seasons, which strongly agree with current MODIS-derived values. Spatially, the hotspot areas are also near the inlets of the major rivers, which are closely linked to the currently delineated hotspot sites.

Second, the NIR reflectance-based retrieval method used in this study has been well tested in optically active inland bodies of water in previous studies. Previous research studies using MODIS have yielded coefficient of determination values (R^2) in the range 0.65–0.85 for satellite-estimated and field-measured TSS and turbidity concentrations, within acceptable error limits for large Lakes and reservoirs (Doxaran *et al* 2002, Shi *et al* 2018). Thirdly, physical consistency checks were carried out by comparing TSS and turbidity values with the mean annual and monthly rainfall. Figure 6 shows the interannual trend of TSS, turbidity, and rainfall. High TSS loads and turbidity are mostly observed in years with high mean annual rainfall.

Monthly rainfall is highly correlated with monthly MODIS-derived TSS and turbidity concentrations. Table 2 describes the monthly rainfall amount range and the derived TSS and turbidity values over Lake Tana

Table 2. Comparison between monthly rainfall and MODIS-derived TSS and turbidity concentrations (2000–2023), illustrating strong coherence between precipitation, suspended sediment dynamics, and turbidity in Lake Tana.

Months	Monthly rainfall (mm month ⁻¹) (2000–2023)	Rainfall pattern	Monthly MODIS-derived TSS (mg l ⁻¹) Turbidity (NTU) from 2000–2023 (current study)	Description
January	0–10	Very low	TSS: 128–288 Turbidity: 89–161	Dry season with no runoff and minimum sediment input to the Lake, and most parts are clear water
February	0–5	Very low	TSS: 102–246 Turbidity: 58–116	Dry month, very low sediment joining the Lake, and clear Lake water
March	5–15	Low	TSS: 105–214 Turbidity: 67–126	Dry month with minimal suspended sediments and very low turbidity
April	15–30	Low	TSS: 186–267 Turbidity: 89–203	Some early rains and a slight rise in turbidity and TSS
May	60–90	Start of rainy season	TSS: 210–325 Turbidity: 116–329	Rain begins, and runoff increases the sediment load to the Lake, and turbidity slightly increases
June	120–200	Rainy season	TSS: 312–497 Turbidity: 187–626	Major rivers' discharge increases, prominent sediment loads to the Lake, and turbidity increases significantly
July	300–380	Peak rains	TSS: 312–619 Turbidity: 398–767	Highest sediment input and peak TSS and turbidity.
August	250–350	Peak rains	TSS: 442–530 Turbidity: 349–501	Strong sediment pulses and high turbidity.
September	150–220	Late Rainy	TSS: 398–518 Turbidity: 225–433	Declining rain and sediment load, minimizing turbidity; wind re-suspension of sediments
October	50–100	End of the rains	TSS: 331–411 Turbidity: 151–326	Post-rain runoff, decreasing sediment, and sediment redistribution by wind and waves
November	15–30	Drying	TSS: 273–346 Turbidity: 120–308	Dry conditions resumed, and the Lake water became clear.
December	0–5	Dry	TSS: 195–263 Turbidity: 106–296	Low suspended solids and low turbidity

4. Conclusions

The use of remote sensing technology enables wide-scale, long-term, and cost-effective monitoring of environmental parameters in large water bodies. In this study, the key open-water quality parameters, TSS and turbidity, were assessed to support sustainable management of water resources in Lake Tana, Northwestern Ethiopia. High-temporal-resolution MODIS satellite imagery was used to track the long-term dynamics of TSS and turbidity and to delineate hotspots. The results show an increasing trend in TSS and turbidity over two decades (from 2000 to 2020), followed by recent declines from 2020 to 2023. The recent decline may indicate reduced rainfall intensity, the recent green legacy in Ethiopia, and good watershed conservation in the upper catchments. Significant variation in TSS and turbidity is observed between wet and dry months. The parameter reaches its peak concentration during the rainy seasons of June, July, and August. This is due to a high sediment load in the Lake from runoff from agricultural upper catchments. In dry months, the TSS concentration is low, and the Lake water is mostly clear. Spatially, the inlets of the major feeder rivers of Lake Tana, including Gilgel Abay, Gumara, Rib, and Megech, exhibited very high concentrations. Those parts of the Lake are determined as hotspot sites. In particular, the Gilgel Abay River is the dominant sediment feeder to the Lake, as the catchment is sloping and has characteristic terrain. The sediments enter the Lake directly, without first being deposited on land. The study supports the sustainable management of Lake waters and provides insights for improving water quality in Ethiopia and globally.

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Data availability statement

The data cannot be made publicly available upon publication because they are owned by a third party and the terms of use prevent public distribution. The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

The author declares no conflicts of interest with any institutions or financial and non-financial matters.

Author contribution

Maru Fentaw Getie  0009-0003-9743-4365

Conceptualization, Data curation, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing

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